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Title

Hospital Patient Admission Analysis

Aim

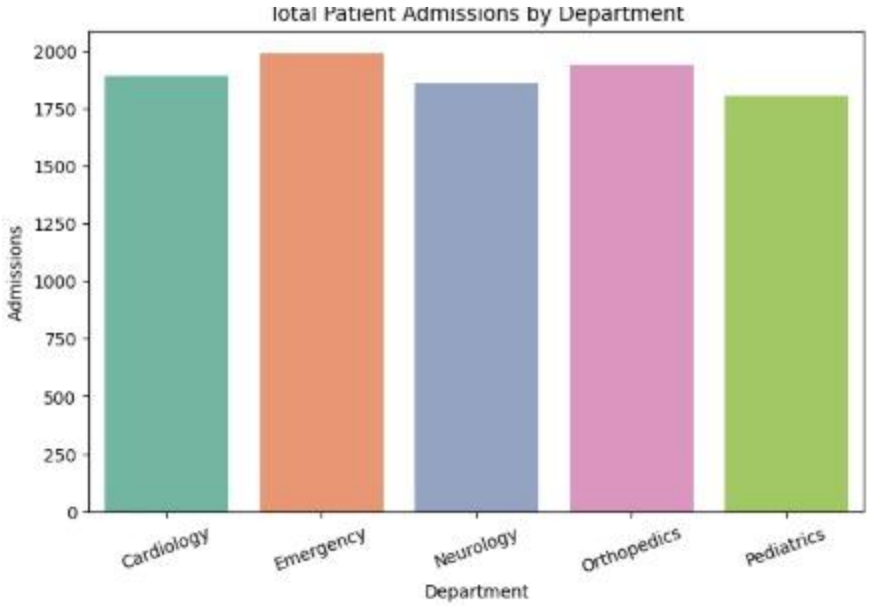
To analyze hospital patient admissions across different departments using data visualization techniques and identify department workload, admission variability, and patient distribution.

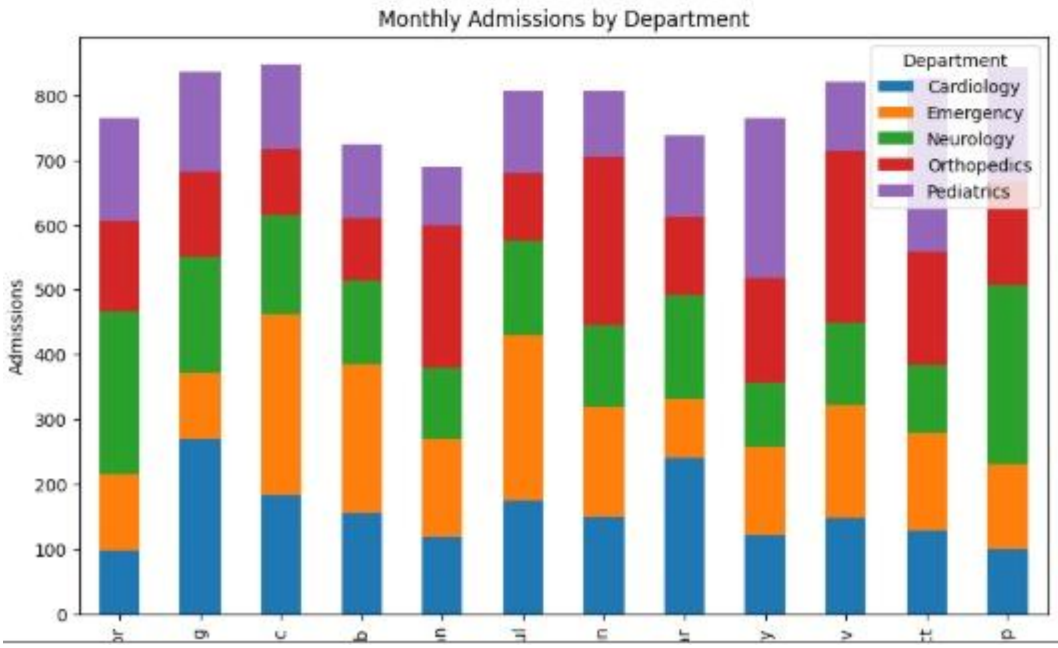
Procedure

1. Open RStudio or Python IDE.
2. Import the required libraries.
3. Create the hospital patient admission dataset.
4. Display the dataset.
5. Generate different visualizations such as Bar Chart, Stacked Bar Chart, Treemap, Box Plot, and Histogram.
6. Compare the visualizations to understand department workload and admission distribution.
7. Analyze the results and identify the most suitable visualization.
8. Interpret the findings.

Python Code

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

import squarify


hospital = pd.DataFrame({
```





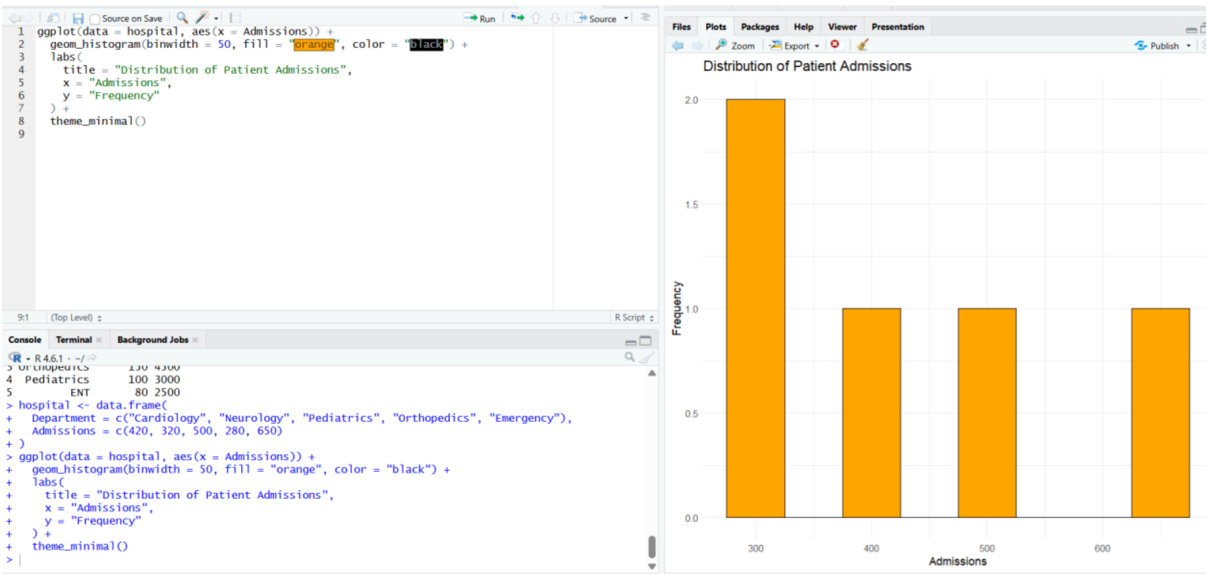

```
    "Department": ["Cardiology","Neurology","Pediatrics","Orthopedics","Emergency"],
    "Admissions": [420,320,500,280,650]
})

print(hospital)

# Bar Chart
sns.barplot(data=hospital, x="Department", y="Admissions")
plt.title("Hospital Patient Admissions")
plt.show()
```

```python
# Histogram
plt.figure()
plt.hist(hospital["Admissions"], bins=5, color="orange", edgecolor="black")
plt.title("Admission Distribution")
plt.xlabel("Admissions")
plt.ylabel("Frequency")
plt.show()

# Box Plot
plt.figure()
sns.boxplot(y=hospital["Admissions"])
plt.title("Admission Variability")
plt.show()

# Treemap
plt.figure(figsize=(7,5))
squarify.plot(
    sizes=hospital["Admissions"],
    label=hospital["Department"],
    alpha=0.8
)
plt.axis("off")
plt.title("Department Workload")
plt.show()
```

R Code

```r
library(ggplot2)
library(treemap)

hospital <- data.frame(
```

```
  Department = c("Cardiology","Neurology","Pediatrics","Orthopedics","Emergency"),
  Admissions = c(420,320,500,280,650)
)

print(hospital)

ggplot(hospital, aes(Department, Admissions, fill=Department)) +
  geom_bar(stat="identity") +
  theme_minimal()

ggplot(hospital, aes(Admissions)) +
  geom_histogram(binwidth=50)

ggplot(hospital, aes(x="", y=Admissions)) +
  geom_boxplot(fill="skyblue")

treemap(
  hospital,
  index="Department",
  vSize="Admissions"
)
```



Output

The program successfully generated different visualizations, including a Bar Chart, Stacked Bar Chart, Treemap, Box Plot, and Histogram. These charts clearly represent department-wise patient admissions, workload, admission variability, and the overall distribution of patients. Among all visualizations, the **Bar Chart** provides the clearest comparison of admissions across departments, while the **Box Plot** highlights admission variability and the **Treemap** effectively shows department workload.

Result

The hospital patient admission data was successfully analyzed using various visualization techniques. The generated graphs helped compare department workloads and patient distributions, making it easier to identify trends and support effective decision-making.

Title

E-Commerce Customer Spending Behaviour

Aim

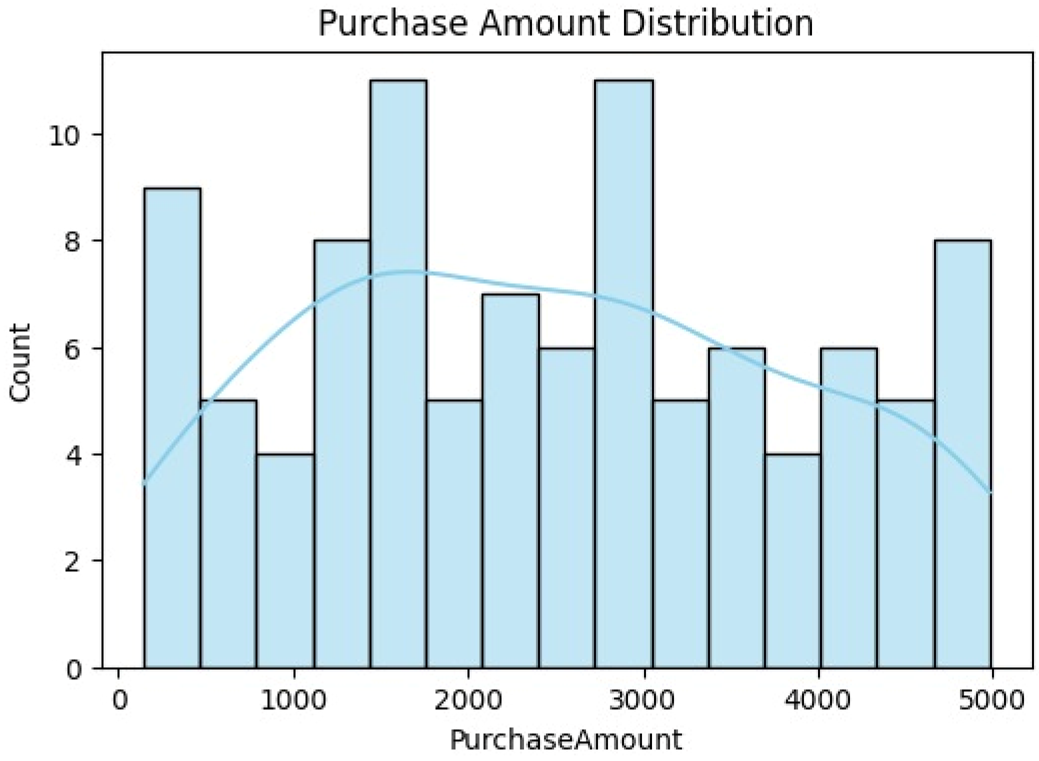
To analyze customer spending behaviour using different visualization techniques and identify high-value customers, spending variability, and purchasing trends.

Procedure

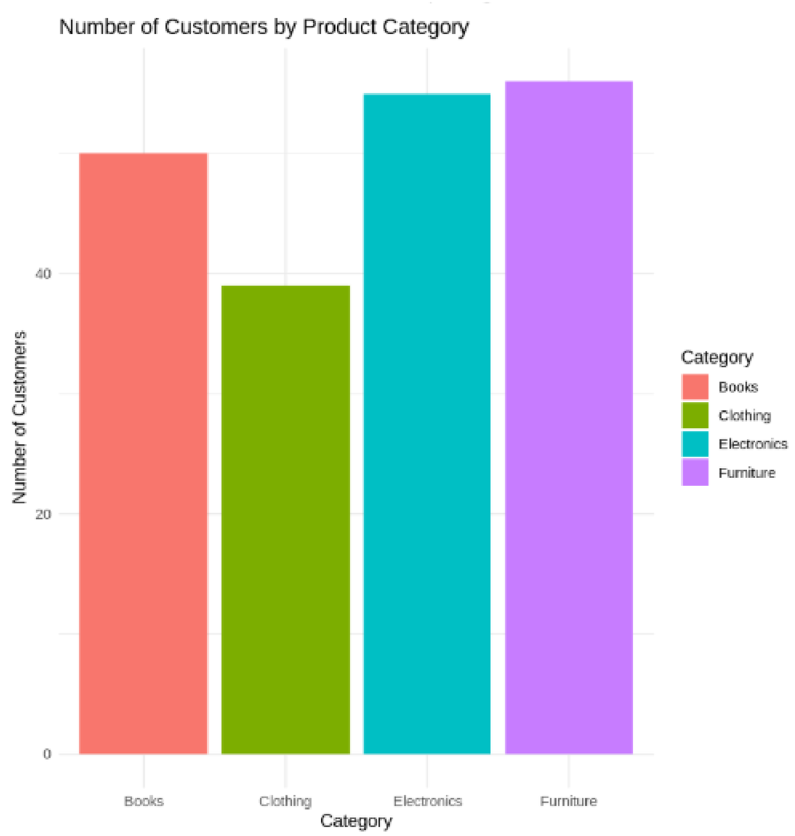
1. Open RStudio or Python IDE.
2. Import the required libraries.
3. Create the customer spending dataset.
4. Display the dataset.
5. Generate visualizations such as Histogram, Violin Plot, Box Plot, Density Plot, and Bar Chart.
6. Compare the visualizations to identify customer spending patterns.
7. Analyze the results and determine the most suitable visualization.
8. Interpret the findings.

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns
```

```
np.random.seed(123)

customer = pd.DataFrame({
    "CustomerID": range(1,201),
    "Spending": np.random.normal(5000,1500,200),
    "OrderFrequency": np.random.randint(1,21,200),
    "Category": np.random.choice(
        ["Electronics","Clothing","Books","Furniture"],200)
})

customer["Spending"] = customer["Spending"].clip(lower=500)

print(customer.head())

# Histogram
plt.figure(figsize=(6,4))
sns.histplot(customer["Spending"], bins=20, color="skyblue")
plt.title("Customer Spending Distribution")
plt.show()

# Violin Plot
plt.figure(figsize=(6,4))
sns.violinplot(data=customer,x="Category",y="Spending")
plt.show()

# Box Plot
```




```
sns.boxplot(data=customer,x="Category",y="Spending")
plt.show()


# Density Plot
plt.figure(figsize=(6,4))
sns.kdeplot(customer["Spending"],fill=True)
plt.show()


# Bar Chart
plt.figure(figsize=(6,4))
sns.countplot(data=customer,x="Category")
plt.show()
```



Output

The program successfully generated a Histogram, Violin Plot, Box Plot, Density Plot, and Bar Chart. These visualizations clearly represent customer spending distribution, purchasing behaviour, and spending variability. The **Violin Plot** and **Box Plot** effectively identify high-value customers and spending variations, while the **Histogram** shows the overall spending distribution.

Result

The customer spending data was successfully analyzed using various visualization techniques. The generated charts helped identify spending patterns, customer behaviour, and purchasing trends, making them useful for business decision-making.

EXPERIMENT 3

Title

University Examination Performance Dashboard

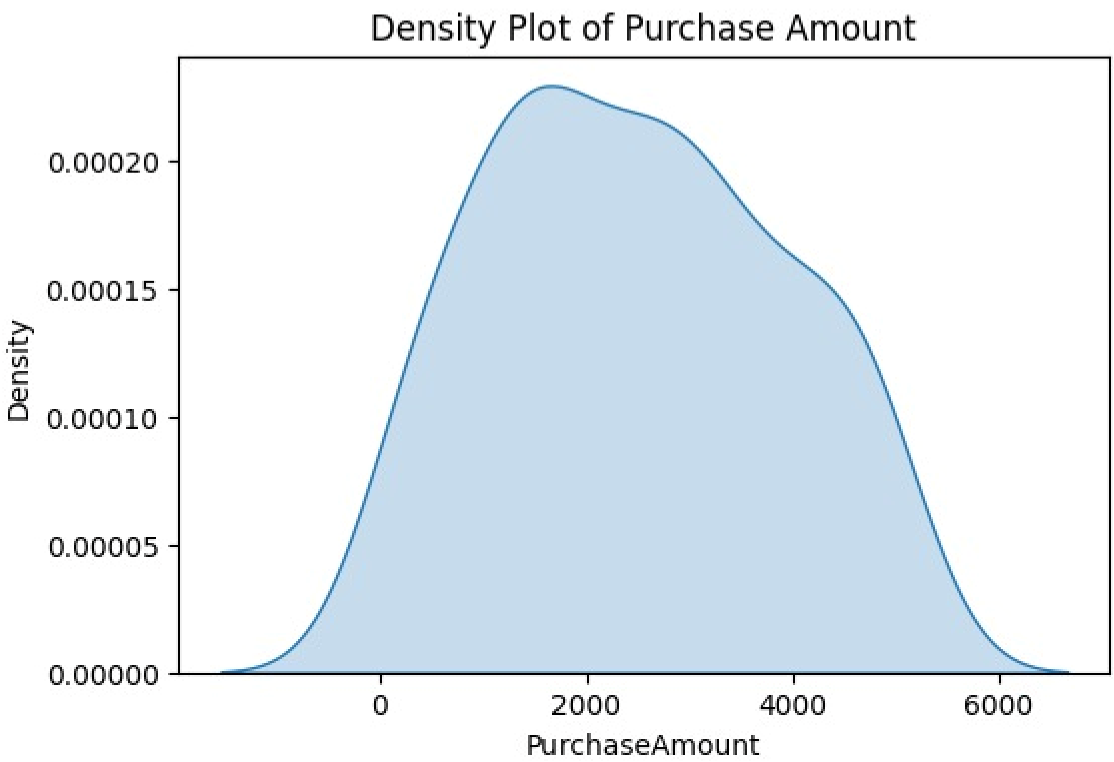
Aim

To analyze student examination performance using different visualization techniques and compare marks across departments to identify score distributions, outliers, and top-performing classes.

Procedure

1. Open RStudio or Python IDE.
2. Import the required libraries.
3. Create the student examination dataset.
4. Display the dataset.
5. Generate visualizations such as Histogram, Box Plot, Violin Plot, Bar Chart, and Scatter Plot.
6. Compare the visualizations to analyze student performance.
7. Identify score distributions, outliers, and department-wise performance.
8. Interpret the results and recommend the best visualization.

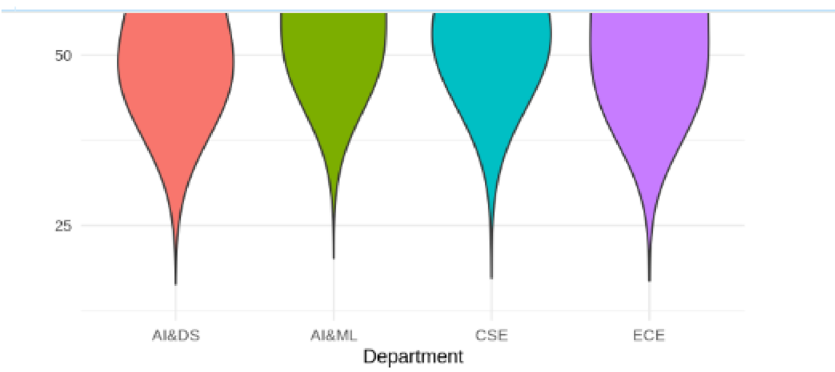
```
library(ggplot2)

set.seed(101)

student <- data.frame(
  StudentID = 1:200,
  Department = sample(c("CSE","ECE","AI&DS","AI&ML"),
                    200,
                    replace=TRUE),
  Marks = sample(40:100,200,replace=TRUE)
)

print(head(student))

# Histogram
ggplot(student,aes(Marks))+
geom_histogram(binwidth=5,fill="skyblue",color="black")+
theme_minimal()
```




```
# Box Plot
ggplot(student,aes(Department,Marks,fill=Department))+
geom_boxplot()+
theme_minimal()

# Violin Plot
ggplot(student,aes(Department,Marks,fill=Department))+
geom_violin()+
theme_minimal()

# Bar Chart
ggplot(student,aes(Department,fill=Department))+
geom_bar()+
theme_minimal()

# Scatter Plot
ggplot(student,aes(Department,Marks,color=Department))+
```

geom_point()+

theme_minimal()



Output

The program successfully generated a Histogram, Box Plot, Violin Plot, Bar Chart, and Scatter Plot. These visualizations clearly represent student marks, score distribution, department-wise performance, and outliers. The **Box Plot** effectively identifies score variations and outliers, while the **Histogram** shows the overall distribution of marks.

Result

The university examination data was successfully analyzed using various visualization techniques. The generated charts provided clear insights into student performance, score distribution, and departmental comparison, helping in effective academic decision-making.

EXPERIMENT 4

Title

Climate and Rainfall Distribution Study

Aim

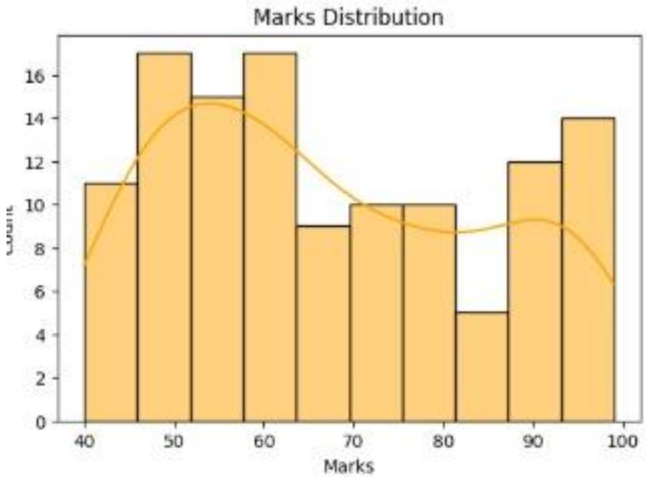
To analyze monthly rainfall, temperature, and humidity data of different cities using various visualization techniques and identify seasonal rainfall patterns and climate distribution.

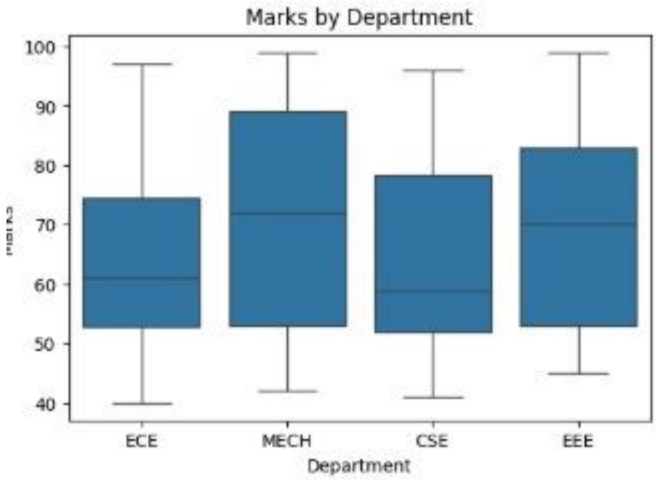
Procedure

1. Open Python IDE or RStudio.
2. Import the required libraries.
3. Create the climate dataset containing city names, rainfall, temperature, and humidity values.
4. Display the dataset.

5. Generate a Histogram to show rainfall distribution.
6. Generate a Density Plot to visualize rainfall trends.
7. Generate a Box Plot to identify rainfall variability and outliers.
8. Generate a Bar Chart to compare rainfall among different cities.
9. Analyze and compare the visualizations.
10. Interpret the results and identify the most suitable visualization for climate analysis.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Create Dataset
climate = pd.DataFrame({
    "City": ["Chennai", "Hyderabad", "Delhi", "Mumbai", "Bangalore"],
    "Rainfall": [120, 90, 60, 210, 140],
    "Temperature": [34, 33, 37, 31, 28],
    "Humidity": [75, 60, 45, 85, 70]
})

print(climate)
```






```
# Histogram
plt.figure(figsize=(6,4))
sns.histplot(climate["Rainfall"], bins=5, color="skyblue")
plt.title("Rainfall Distribution")
plt.show()

# Density Plot
plt.figure(figsize=(6,4))
sns.kdeplot(climate["Rainfall"], fill=True)
plt.title("Density Plot of Rainfall")
plt.show()

# Box Plot
plt.figure(figsize=(6,4))
```

```
sns.boxplot(y=climate["Rainfall"], color="orange")
plt.title("Rainfall Variability")
plt.show()

# Bar Chart
plt.figure(figsize=(6,4))
sns.barplot(data=climate, x="City", y="Rainfall")
plt.title("Rainfall by City")
plt.show()
library(ggplot2)

# Create Dataset
climate <- data.frame(
  City = c("Chennai", "Hyderabad", "Delhi", "Mumbai", "Bangalore"),
  Rainfall = c(120, 90, 60, 210, 140),
  Temperature = c(34, 33, 37, 31, 28),
  Humidity = c(75, 60, 45, 85, 70)
)

print(climate)

# Histogram
ggplot(climate, aes(x = Rainfall)) +
  geom_histogram(binwidth = 30, fill = "skyblue", color = "black") +
  theme_minimal()

# Density Plot
ggplot(climate, aes(x = Rainfall)) +
  geom_density(fill = "lightgreen", alpha = 0.5) +
  theme_minimal()
```

Box Plot

ggplot(climate, aes(x = "", y = Rainfall)) +

  geom_boxplot(fill = "orange") +

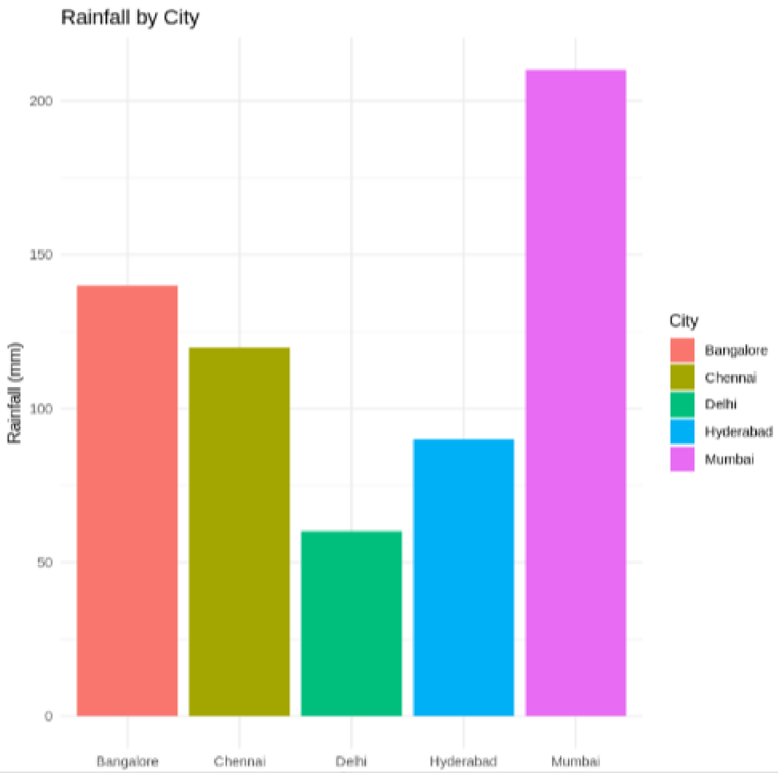
  theme_minimal()

Bar Chart

ggplot(climate, aes(x = City, y = Rainfall, fill = City)) +

  geom_bar(stat = "identity") +

  theme_minimal()



Output

The program successfully generated a Histogram, Density Plot, Box Plot, and Bar Chart. These visualizations clearly represent rainfall distribution, seasonal rainfall variation, and city-wise climate comparison. The Histogram displays the frequency of rainfall values, the Density Plot illustrates rainfall trends, the Box Plot highlights variability and possible outliers, and the Bar Chart compares rainfall among different cities.

Result

The climate and rainfall data was successfully analyzed using different visualization techniques. The generated graphs provided clear insights into rainfall distribution and climate variations, making them useful for environmental research and weather analysis.

EXPERIMENT 5

Title

Bank Loan Risk Assessment

Aim

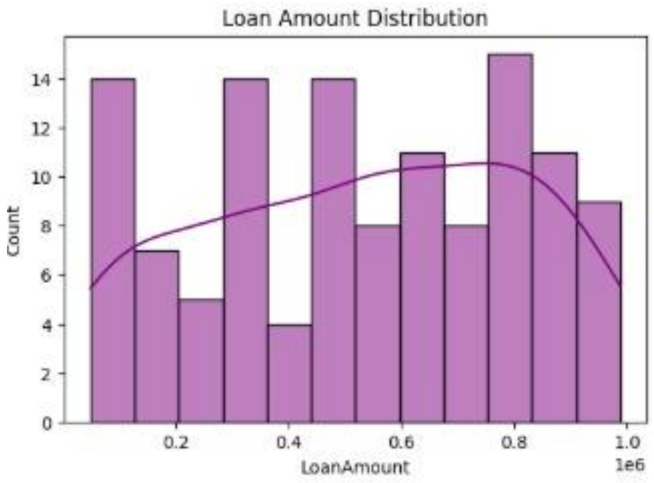
To analyze customer loan data using visualization techniques and identify loan distribution, repayment behaviour, credit score variation, and customer risk patterns.

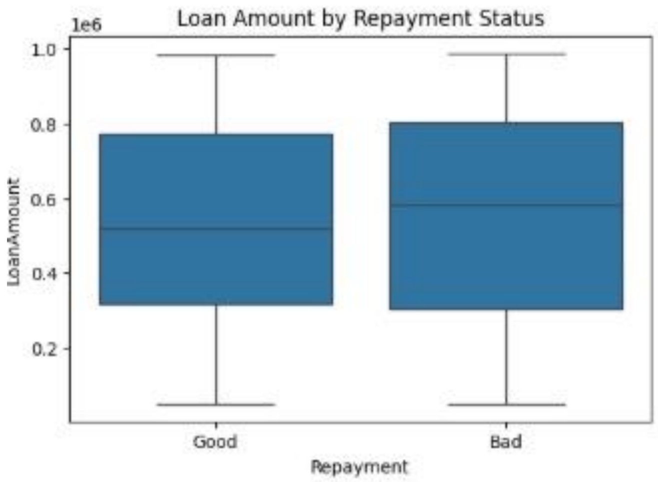
Procedure

1. Open Python IDE or RStudio.
2. Import the required libraries.
3. Create the bank loan dataset containing customer income, loan amount, credit score, and repayment status.
4. Display the dataset.
5. Generate a Histogram to visualize loan amount distribution.
6. Generate a Scatter Plot to compare income and loan amount.
7. Generate a Box Plot to analyze credit score variations based on repayment status.
8. Generate a Bar Chart to compare customer repayment categories.
9. Analyze the generated visualizations.
10. Interpret the results and identify the most suitable visualization for loan risk assessment.

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns


np.random.seed(111)


loan = pd.DataFrame({

    "CustomerID": range(1,201),
```

```
    "Income": np.random.normal(50000,12000,200),
    "LoanAmount": np.random.normal(250000,60000,200),
    "CreditScore": np.random.randint(300,901,200),
    "Repayment": np.random.choice(["Good","Average","Poor"],200)
})

loan["Income"] = loan["Income"].clip(lower=10000)
loan["LoanAmount"] = loan["LoanAmount"].clip(lower=50000)

print(loan.head())

# Histogram
plt.figure(figsize=(6,4))
sns.histplot(loan["LoanAmount"], bins=20, color="skyblue")
plt.title("Loan Amount Distribution")
plt.show()

# Scatter Plot
plt.figure(figsize=(6,4))
sns.scatterplot(data=loan, x="Income", y="LoanAmount", hue="Repayment")
plt.title("Income vs Loan Amount")
plt.show()

# Box Plot
plt.figure(figsize=(6,4))
sns.boxplot(data=loan, x="Repayment", y="CreditScore")
```

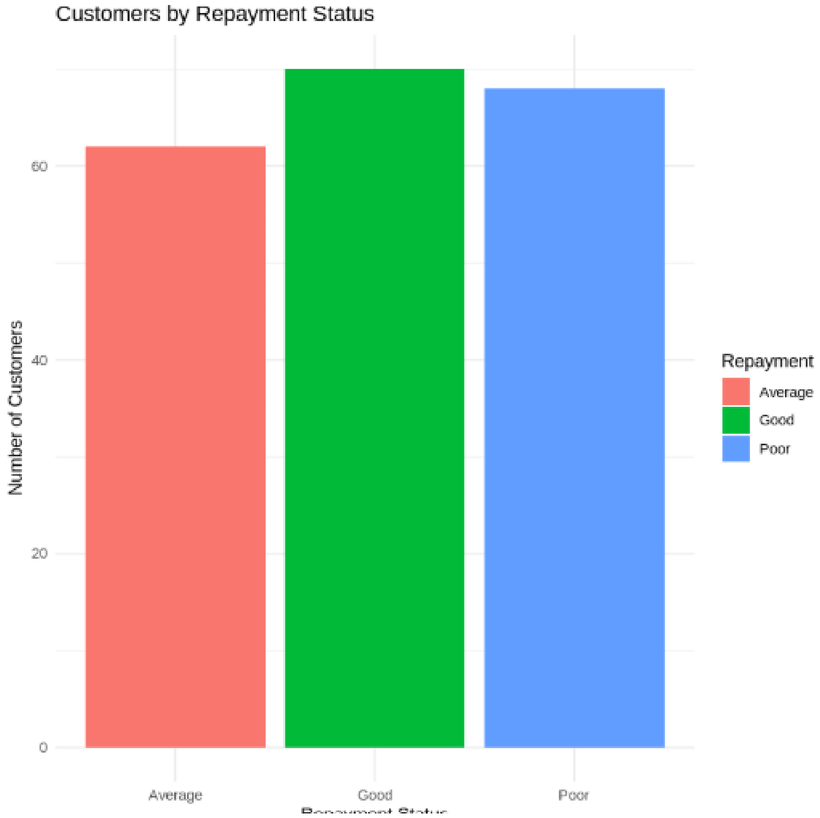





```
plt.title("Credit Score by Repayment Status")
plt.show()

# Bar Chart
plt.figure(figsize=(6,4))
sns.countplot(data=loan, x="Repayment")
plt.title("Customers by Repayment Status")
plt.show()

library(ggplot2)

set.seed(111)

loan <- data.frame(
```

```r
  CustomerID = 1:200,
  Income = round(rnorm(200, 50000, 12000)),
  LoanAmount = round(rnorm(200, 250000, 60000)),
  CreditScore = sample(300:900, 200, replace = TRUE),
  Repayment = sample(c("Good", "Average", "Poor"), 200, replace = TRUE)
)

loan$Income[loan$Income < 10000] <- 10000
loan$LoanAmount[loan$LoanAmount < 50000] <- 50000

print(head(loan))

# Histogram
ggplot(loan, aes(x = LoanAmount)) +
  geom_histogram(binwidth = 20000, fill = "skyblue", color = "black") +
  theme_minimal()

# Scatter Plot
ggplot(loan, aes(x = Income, y = LoanAmount, color = Repayment)) +
  geom_point(size = 2) +
  theme_minimal()

# Box Plot
ggplot(loan, aes(x = Repayment, y = CreditScore, fill = Repayment)) +
  geom_boxplot() +
  theme_minimal()

# Bar Chart
ggplot(loan, aes(x = Repayment, fill = Repayment)) +
  geom_bar() +
  theme_minimal()
```

Output



The program successfully generated a Histogram, Scatter Plot, Box Plot, and Bar Chart. These visualizations clearly represent loan amount distribution, customer income, repayment behaviour, and credit score variations. The Scatter Plot shows the relationship between customer income and loan amount, while the Box Plot highlights credit score differences among repayment categories. The Histogram and Bar Chart provide a clear overview of loan distribution and repayment status.

Result

The bank loan data was successfully analyzed using various visualization techniques. The generated charts helped identify loan risk patterns, customer repayment behaviour, and credit score distribution, making them useful for financial institutions in loan approval and risk assessment.